Vimansh Mahajan

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LinkedIn | Portfolio | GitHub

Education

Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi) CGPA: 9.28 / 10.00
Bachelor of Technology (B.Tech.) in Computer Science Engineering (CSE) *2022 – 2026*
- Dean's List (2024–25) • Flipkart GRiD 7.0 National Semi-Finalist • Adobe GenSolve Top 5%

The Vivekanand School, Narela, New Delhi CBSE Class XII: 97.2%
- JEE Main 2022: 99 Percentile (AIR 9,200) • JEE Advanced 2022: AIR 10,109

Experience

Bank of New York (BNY) Aug 2026 – Present
Engineering Analyst – Software Engineering
- Contributing to software engineering and application configuration changes within the **Structured Debt Platform**, working with the team to understand and modify existing systems.

Bank of New York (BNY) May 2025 – July 2025
Software Engineer Intern
- Architected a production-grade content automation system using **Spring Boot**, **Python LLM services**, **Oracle DB**, and **Angular**, transforming raw finance notes and images into publication-ready market commentary.
- Reduced content turnaround time from **4–5 hours to under 10 minutes** by automating structured note capture, AI drafting in author style, contextual image insertion, and final publishing workflows.
- Containerized backend services and deployed production workloads on **Google App Engine**, improving portability and release efficiency.

Complex Systems (CoSy) Lab Jan 2025 – May 2025
UG Researcher
- Developed **RecipeGPT**, a generative NLP system trained on the Recipe1M corpus, delivering high-quality recipe generation with **BERTScore** $>$ **0.85** and **ROUGE-L 0.32**.
- Built a custom **NER-driven preprocessing pipeline** to normalize ingredient inputs and improve model conditioning on structured entities.
- Designed a **3-stage GPT-2 fine-tuning pipeline** over **100k+ recipe instructions**, progressively training on instructions, ingredient-title conditioning, and NER-enhanced tokenized inputs with label masking.

Projects

OptiWealth: Portfolio Optimization Platform | *Spring Boot, Python, yfinance, React, PostgreSQL* Nov 2025
- Architected a multi-service portfolio analytics platform with **React** frontend, **Spring Boot REST** backend, and Python compute services for quantitative workloads.
- Built JWT-authenticated portfolio workflows using **Spring Security** for creation, tracking, and optimization.
- Implemented portfolio analytics covering returns, volatility, Sharpe ratio, **VaR/CVaR**, beta, drawdown, and diversification across NSE-listed equities.
- Developed forecasting and optimization engines using **ARIMA**, **GARCH**, **Monte Carlo simulation**, efficient frontier, and max-Sharpe/min-volatility strategies, with AI-generated plain-language investment insights.

OncoHelp-AI: Brain MRI Decision Support | *Python, Streamlit, Grad-CAM, LLMs* Dec 2025
- Architected a multi-stage **brain MRI decision-support system** using ResNet18 and EfficientNet models for image validation, tumor screening, and subtype classification.
- Integrated **Grad-CAM heatmaps** and **Monte Carlo Dropout**–based uncertainty estimation to surface interpretable predictions with calibrated confidence signals.
- Developed a safety-constrained LLM assistant using Gemini and BioGPT for conservative summaries and follow-ups.

Skills

Languages: Python, Java, C++
Frameworks & Libraries: Spring Boot, React, Angular, Hugging Face Transformers, PyTorch
Databases & Tools: PostgreSQL, MySQL, Oracle DB, Git, Docker, REST APIs
Core CS: Data Structures & Algorithms, OOP, DBMS, Computer Networks, System Design
AI / ML: Machine Learning, NLP, LLM Fine-Tuning, Model Evaluation, Data Science

Additional Information

Coding Platforms: Leetcode (300+ problems solved), Codeforces
Certification: Quantitative Finance & Algorithmic Trading in Python
Hobbies: Debate, Fitness, Badminton
Volunteering: Volunteer, Seva Sankalp NGO